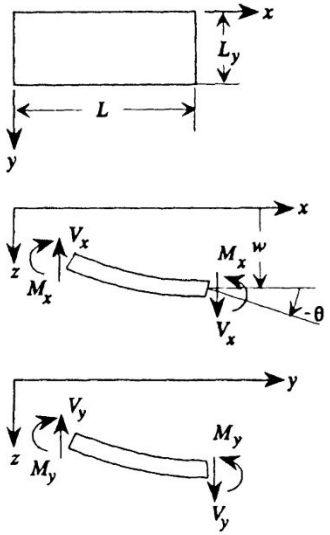
TABLE 18-16 DEFLECTIONS AND INTERNAL FORCES OF RECTANGULAR PLATES

Notation

w = deflection
E = modulus of elasticity
ν = Poisson's ratio
h = thickness of plate
$D = Eh^3/[12(1-\nu^2)]$
W_T = concentrated loading (F)
p_1 = uniformly distributed loading (F/L^2)
$\alpha = L_y/L$
$\beta = L/L_y$
M_x, M_y = bending moment per unit length on surface normal to x and y axes
Q_x, Q_y = shear force per unit length on surfaces normal to x and y axes
V_x, V_y = equivalent shear force per unit length acting on planes normal to x and y axes



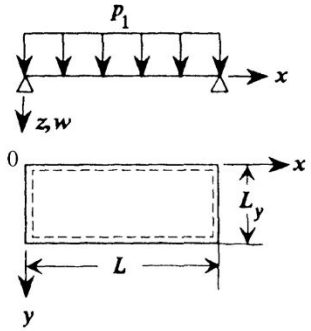
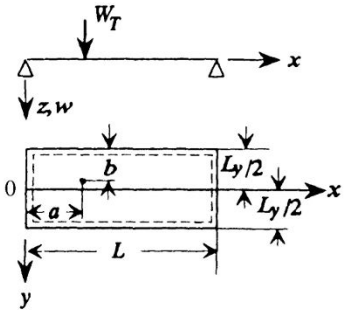
Structural System and Static Loading	Deflection and Internal Forces
1. Simply supported on all edges, uniform loading 	$w = \dfrac{16p_1L^4}{\pi^6 D}\sum_m\sum_n \dfrac{\sin(n\pi x/L)\sin(m\pi y/L_y)}{mn(n^2/L^2+m^2/L_y^2)^2}$ $M_x = \dfrac{16p_1L^2}{\pi^4}\sum_m\sum_n \gamma_1\dfrac{\sin(n\pi x/L)\sin(m\pi y/L_y)}{mn(n^2+m^2/\alpha^2)^2}$ $M_y = \dfrac{16p_1L^2}{\pi^4}\sum_m\sum_n \gamma_2\dfrac{\sin(n\pi x/L)\sin(m\pi y/L_y)}{mn(n^2+m^2/\alpha^2)^2}$ $w_{\max} = c_1\dfrac{p_1L^4}{Eh^3} = w_{\text{center}}$ $\gamma_1 = (n^2+\nu m^2/\alpha^2), \qquad \gamma_2 = (m^2/\alpha^2+\nu n^2)$

TABLE 18-16 (continued) DEFLECTIONS AND INTERNAL FORCES OF RECTANGULAR PLATES

Structural System and Static Loading	Deflection and Internal Forces
	$(M_x)_{\max} = c_2 p_1 L^2$ (center) $(M_y)_{\max} = c_3 p_2 L^2$ (center) $(Q_x)_{\max} = c_4 p_1 L$ ($x = 0$, L edges) $(Q_y)_{\max} = c_5 p_1 L_y$ ($y = 0$, L_y edges) $(V_x)_{\max} = c_6 p_1 L$ (lateral edge forces) $(V_y)_{\max} = c_7 p_1 L_y$ (lateral edge forces) $R_0 = c_8 p_1 L L_y$ (corner forces) CONSTANTS: $m = 1, 3, 5, \ldots, \infty$ $\quad n = 1, 3, 5, \ldots, \infty$ $\nu = 0.3$ $c_1 = 0.1421 + 0.08204\beta - 0.3985\beta^2 + 0.219\beta^3$ $c_2 = 0.1247 + 0.05524\beta - 0.2762\beta^2 + 0.1441\beta^3$ $c_3 = 0.03726 - 0.01504\beta + 0.1028\beta^2 - 0.07756\beta^3$ $c_4 = 0.4967 + 0.1686\beta - 0.6849\beta^2 + 0.3976\beta^3$ $c_5 = -0.0007805 + 0.3679\beta + 0.04574\beta^2 - 0.07488\beta^3$ $c_6 = 0.4983 + 0.02879\beta + 0.004425\beta^2 - 0.1137\beta^3$ $c_7 = -0.003591 + 0.5091\beta + 0.05874\beta^2 - 0.1445\beta^3$ $c_8 = -0.003545 + 0.1179\beta - 0.02755\beta^2 - 0.02202\beta^3$

2.

Simply supported on all edges, concentrated load



$$w = \frac{4W_T}{D\pi^4 L L_y} \sum_m \sum_n \frac{\sin(n\pi a/L)\sin(m\pi b/L_y)\sin(n\pi x/L)\sin(m\pi y/L_y)}{(n^2/L^2 + m^2/L_y^2)^2}$$

$m = 1, 2, 3, \ldots, ; n = 1, 2, 3, \ldots$

or

$$w = \frac{W_T L^2}{D\pi^3} \sum_{n=1}^{\infty} \left(1 + \beta_n \coth \beta_n - \frac{\beta_n y_1}{L_y} \coth \frac{\beta_n y_1}{L_y} - \frac{\beta_n b_1}{L_y} \coth \frac{\beta_n b_1}{L_y}\right) \times \frac{\sinh \frac{\beta_n b_1}{L_y} \sinh \frac{\beta_n y_1}{L_y} \sin \frac{n\pi a}{L} \sin \frac{m\pi x}{L}}{n^3 \sinh \beta_n}$$

If $y \geq b$, use $y_1 = L_y - y$ and $b_1 = b$

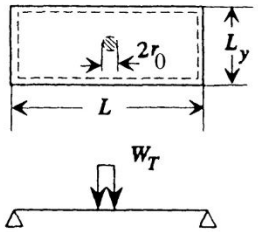
If $y < b$, use $y_1 = y$ and $b_1 = L_y - b$

$$\beta_n = \frac{n\pi L_y}{L}$$

$n = 1, 2, 3, \ldots$

3.

Simply supported on all edges, uniform load on a small circle of radius r_0 at center of plate



At center:

$$w_{\max} = k_1 \frac{W_T L_y^2}{E h^3}$$

$$\sigma_{\max} = \frac{3W_T}{2\pi h^2}\left[(1+\nu)\ln\frac{2L_y}{\pi r_e} + k_2\right]$$

where

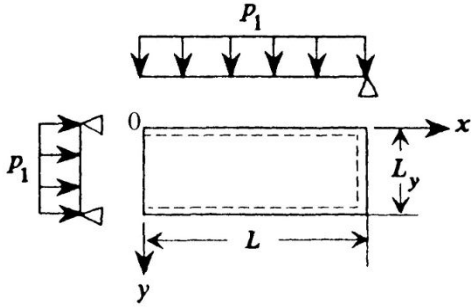
$$r_e = \begin{cases} \sqrt{1.6r_0^2 + h^2} - 0.675h & r_0 < 0.5h \\ r_0 & r_0 > 0.5h \end{cases}$$

W_T = total load on plate

$k_1 = 0.1851 + 0.06342\alpha - 0.1643\alpha^2 + 0.04232\alpha^3$

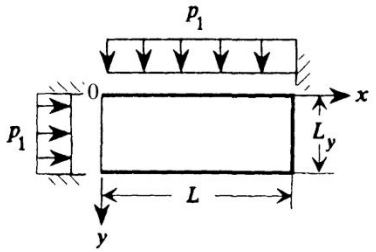
$k_2 = 0.9998 + 0.5195\alpha - 1.29\alpha^2 + 0.2042\alpha^3$

TABLE 18-16 (continued) DEFLECTIONS AND INTERNAL FORCES OF RECTANGULAR PLATES

Structural System and Static Loading	Deflection and Internal Forces
4. Three edges simply supported, one edge free, uniform load  	$(M_x)_{\substack{x=L/2 \\ y=L_y/2}} = c_1 p_1 L_y^2$ $(M_y)_{\substack{x=L/2 \\ y=L_y/2}} = c_2 p_1 L_y^2$ $(M_y)_{\substack{x=0 \\ y=L_y/2}} = c_3 p_1 L_y^2$ $\nu = 0.15$ CONSTANTS: $c_1 = \begin{cases} -2.7975 + 16.537\beta^{1/2} - 36.423\beta + 35.523\beta^{3/2} - 12.913\beta^2 & 0.3 \le \beta < 0.8 \\ -0.2714 + 0.6371\beta^{1/2} - 0.4533\beta + 0.1021\beta^{3/2} & 0.8 \le \beta \le 2.0 \end{cases}$ $c_2 = 0.1448 - 0.7829\beta^{1/2} + 1.415\beta - 0.9113\beta^{3/2} + 0.2039\beta^2 \qquad 0.3 \le \beta \le 2.0$ $c_3 = 0.2558 - 1.3186\beta^{1/2} + 2.6532\beta - 1.9313\beta^{3/2} + 0.4817\beta^2 \qquad 0.3 \le \beta \le 2.0$

5.

Three edges clamped, one edge free, uniform load



$$(M_x)_{\substack{x=L/2\\y=L_y/2}} = c_1 p_1 L_y^2$$

$$(M_y)_{\substack{x=L/2\\y=L_y/2}} = c_2 p_1 L_y^2$$

$$(M_y)_{\substack{x=0\\y=L_y/2}} = c_3 p_1 L_y^2$$

$$(M_x)_{\substack{x=L\\y=L_y/2}} = c_4 p_1 L^2$$

$$(M_y)_{\substack{x=L/2\\y=0,L_y}} = c_5 p_1 L_y^2$$

$$(M_y)_{\substack{x=0\\y=0,L_y}} = c_6 p_1 L_y^2$$

$\nu = 0.15$

CONSTANTS:

$$c_1 = \begin{cases} -0.01247 + 0.05532\alpha - 0.04778\alpha^2 + 0.01519\alpha^3 - 0.001687\alpha^4 & 1.25 < \alpha \le 3.3 \\ -0.006967 + 0.01615\alpha + 0.01147\alpha^2 - 0.01205\alpha^3 & 0.5 \le \alpha \le 1.25 \end{cases}$$

$$c_2 = 0.05474 - 0.03243\alpha + 0.006359\alpha^2 - 0.0003985\alpha^3 \qquad 0.5 \le \alpha \le 3.3$$

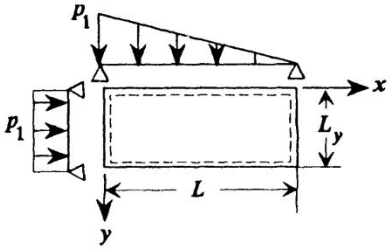
$$c_3 = 0.03724 + 0.02528\alpha - 0.02263\alpha^2 + 0.003719\alpha^3 \qquad 0.5 \le \alpha \le 3.3$$

$$c_4 = -0.006349 + 0.02708\alpha - 0.09165\alpha^2 + 0.01488\alpha^3 \qquad 0.5 \le \alpha \le 3.3$$

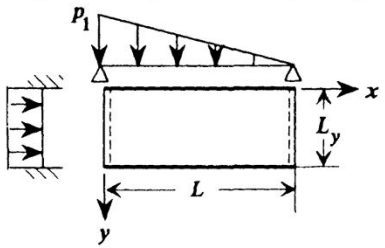
$$c_5 = -0.1062 + 0.04391\alpha - 0.00235\alpha^2 - 0.0007418\alpha^3 \qquad 0.5 \le \alpha \le 3.3$$

$$c_6 = -0.06827 - 0.04205\alpha + 0.02756\alpha^2 - 0.003528\alpha^3 \qquad 0.5 \le \alpha \le 3.3$$

TABLE 18-16 (continued) DEFLECTIONS AND INTERNAL FORCES OF RECTANGULAR PLATES

Structural System and Static Loading	Deflection and Internal Forces
6. Simply supported on all edges, linearly varying load 	If $L/L_y < 1.00$, $(M_x)_{\substack{x=L/2\\y=L_y/2}} = c_1 p_1 L^2$ $(M_x)_{\max} = c_2 p_1 L^2$ $(M_y)_{\substack{x=L/2\\y=L_y/2}} = c_3 p_1 L^2$ $(M_y)_{\max} = c_4 p_1 L^2$ If $L/L_y \geq 1.00$, replace $p_1 L^2$ by $p_1 L_y^2$. $\nu = 0.15$ CONSTANTS: $c_1 = \begin{cases} 0.08438 - 0.06242\beta - 0.03592\beta^2 + 0.03239\beta^3 & 0.5 \leq \beta \leq 1.0 \\ -0.003636 + 0.05639\beta - 0.04352\beta^2 + 0.009206\beta^3 & 1.0 < \beta \leq 2.0 \end{cases}$ $c_2 = \begin{cases} 0.085096 - 0.0635\beta - 0.02495\beta^2 + 0.02498\beta^3 & 0.5 \leq \beta \leq 1.0 \\ -0.01872 + 0.07831\beta - 0.04657\beta^2 + 0.008649\beta^3 & 1.0 < \beta \leq 2.0 \end{cases}$ $c_3 = \begin{cases} 0.01873 + 0.07069\beta - 0.02985\beta^2 + 0.003716\beta^3 & 0.5 < \beta \leq 1.0 \\ -0.04394 + 0.08744\beta - 0.03008\beta^2 + 0.004946\beta^3 & 1.0 < \beta \leq 2.0 \end{cases}$ $c_4 = \begin{cases} 0.01756 - 0.05004\beta - 0.1028\beta^2 + 0.05186\beta^3 & 0.5 \leq \beta \leq 1.0 \\ -0.01646 + 0.02599\beta + 0.01385\beta^2 + 0.004938\beta^3 & 1.0 < \beta \leq 2.0 \end{cases}$

7.
Two edges simply supported, two edges clamped, linearly varying load



If $\beta = L/L_y < 1.00$,

$$-(M_y)_{\max} = c_1 p_1 L^2$$

$$(M_x)_{\substack{x=L/2 \\ y=0,L_y}} = c_2 p_1 L^2$$

$$(M_x)_{\max} = c_3 p_1 L^2$$

$$(M_x)_{\substack{x=L/2 \\ y=L_y/2}} = c_4 p_1 L^2$$

$$(M_y)_{\max} = c_5 p_1 L^2$$

$$(M_y)_{\substack{x=L/2 \\ y=L_y/2}} = c_6 p_1 L^2$$

If $\beta \geq 1.00$, replace $p_1 L^2$ by $p_1 L_y^2$.

$\nu = 0.15$

CONSTANTS:

$0.5 \leq \beta \leq 1.0$

$$c_1 = -0.0947 + 0.09607\beta - 0.0758\beta^2 + 0.03702\beta^3$$

$$c_2 = -0.07833 + 0.01816\beta + 0.04384\beta^2 - 0.01853\beta^3$$

$$c_3 = 0.147 - 0.3292\beta + 0.286\beta^2 - 0.09076\beta^3$$

$$c_4 = 0.120 - 0.218\beta + 0.1254\beta^2 - 0.01945\beta^3$$

$$c_5 = -0.04249 + 0.196\beta - 0.2114\beta^2 + 0.07222\beta^3$$

$$c_6 = -0.04249 + 0.196\beta - 0.2114\beta^2 + 0.07222\beta^3$$

$1.0 < \beta \leq 2.0$

$$c_1 = -0.001352 - 0.04739\beta + 0.01262\beta^2 - 0.001445\beta^3$$

$$c_2 = 0.01942 - 0.09533\beta + 0.04973\beta^2 - 0.008762\beta^3$$

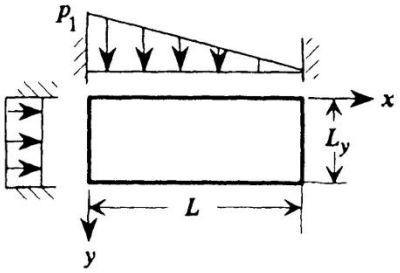
$$c_3 = -0.005903 + 0.04346\beta - 0.03065\beta^2 + 0.00615\beta^3$$

$$c_4 = 0.02255 - 0.01713\beta + 0.001814\beta^2 + 0.0006693\beta^3$$

$$c_5 = -0.0156 + 0.04088\beta - 0.01164\beta^2 + 0.00638\beta^3$$

$$c_6 = -0.02249 + 0.06271\beta - 0.03123\beta^2 + 0.005338\beta^3$$

TABLE 18-16 (continued) DEFLECTIONS AND INTERNAL FORCES OF RECTANGULAR PLATES

Structural System and Static Loading	Deflection and Internal Forces
8. Clamped on all edges, linearly on varying load 	If $\beta = L/L_y < 1.00$, $(M_x)_{\substack{x=0 \\ y=L_y/2}} = c_1 p_1 L^2$ $(M_x)_{\substack{x=L \\ y=L_y/2}} = c_2 p_1 L^2$ $-(M_y)_{\max} = c_3 p_1 L^2$ $(M_y)_{\substack{x=L/2 \\ y=0,L_y}} = c_4 p_1 L^2$ $(M_x)_{\max} = c_5 p_1 L^2$ $(M_x)_{\substack{x=L/2 \\ y=L_y/2}} = c_6 p_1 L^2$ $(M_y)_{\max} = c_7 p_1 L^2$ $(M_y)_{\substack{x=L/2 \\ y=L_y/2}} = c_8 p_1 L^2$ If $\beta \geq 1.00$, replace $p_1 L$ by $p_1 L_y^2$. $\nu = 0.15$

CONSTANTS:

$0.5 \le \beta \le 1.0$

$c_1 = -0.03596 - 0.08083\beta + 0.1319\beta^2 - 0.04816\beta^3$

$c_2 = -0.01030 - 0.116\beta + 0.1791\beta^2 - 0.07038\beta^3$

$c_3 = -0.02636 - 0.004588\beta - 0.009978\beta^2 + 0.01388\beta^3$

$c_4 = -0.02993 + 0.01295\beta - 0.02866\beta^2 + 0.02036\beta^3$

$c_5 = 0.0261 - 0.00459\beta - 0.01879\beta^2 + 0.007399\beta^3$

$c_6 = 0.01218 + 0.05503\beta - 0.09913\beta^2 + 0.04073\beta^3$

$c_7 = 0.03321 - 0.1267\beta + 0.1792\beta^2 - 0.07687\beta^3$

$c_8 = -0.005465 + 0.003079\beta + 0.03526\beta^2 - 0.02409\beta^3$

$1.0 < \beta \le 2.0$

$c_1 = 0.02841 - 0.1064\beta + 0.0549\beta^2 - 0.01001\beta^3$

$c_2 = 0.005924 - 0.05511\beta + 0.03986\beta^2 - 0.00827\beta^3$

$c_3 = 0.03386 - 0.08143\beta + 0.02145\beta^2 - 0.0008475\beta^3$

$c_4 = 0.04529 - 0.1128\beta + 0.04949\beta^2 - 0.007329\beta^3$

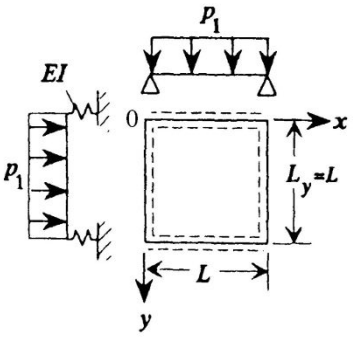
$c_5 = -0.00482 + 0.03546\beta - 0.02663\beta^2 + 0.006204\beta^3$

$c_6 = -0.00233 + 0.03325\beta - 0.02862\beta^2 + 0.006528\beta^3$

$c_7 = -0.01682 + 0.02965\beta - 0.002959\beta^2 - 0.001043\beta^3$

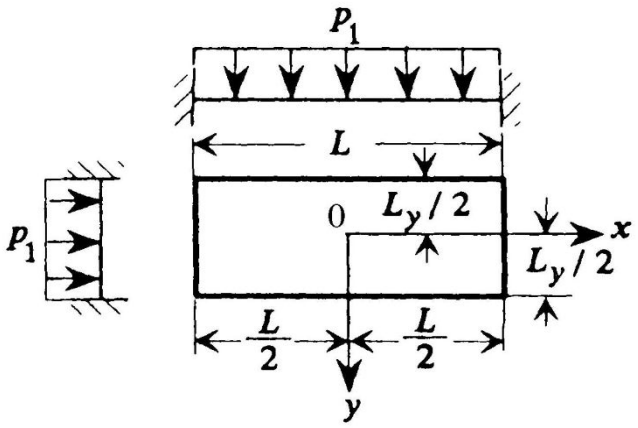
$c_8 = -0.03277 + 0.06381\beta - 0.02579\beta^2 + 0.003542\beta^3$

TABLE 18-16 (continued) DEFLECTIONS AND INTERNAL FORCES OF RECTANGULAR PLATES

Structural System and Static Loading	Deflection and Internal Forces
9. Two edges simply supported, two edges flexibly supported, uniform loading  	$w_{max} = c_1 \dfrac{p_1 L^4}{D} = w_{center}$ $(M_x)_{max} = c_2 p_1 L^2$ (center) $(M_y)_{max} = c_3 p_1 L^2$ (center) $\nu = 0.3$ EI = stiffness of edge beam supports CONSTANTS: $c_1 = 0.004037 + 0.003975\dfrac{1}{1+\eta} - 0.003862\dfrac{1}{(1+\eta)^2} + 0.008931\dfrac{1}{(1+\eta)^3}$ $c_2 = 0.04756 + 0.0356\dfrac{1}{1+\eta} - 0.05788\dfrac{1}{(1+\eta)^2} + 0.1001\dfrac{1}{(1+\eta)^3}$ $c_3 = 0.04759 - 0.00504\dfrac{1}{1+\eta} - 0.05026\dfrac{1}{(1+\eta)^2} + 0.03471\dfrac{1}{(1+\eta)^3}$ $\eta = \dfrac{EI}{LD} \geq 0$

10.

Clamped on all edges uniform loading



$$w_{\max} = c_1 \frac{p_1 L^4}{E h^3} = w_{\text{center}}$$

$$(M_x)_{\substack{x=0 \\ y=0}} = c_2 p_1 L^2 \qquad (M_y)_{\substack{x=0 \\ y=0}} = c_3 p_1 L^2$$

$$(M_x)_{\substack{x=L/2 \\ y=0}} = -c_4 p_1 L^2 \qquad (M_y)_{\substack{x=0 \\ y=L_y/2}} = -c_5 p_1 L^2$$

$\nu = 0.3$

CONSTANTS:

$1.0 \le \alpha \le 2.0$

$$c_1 = -0.06479 + 0.1327\alpha - 0.0665\alpha^2 + 0.01162\alpha^3$$

$$c_2 = -0.07859 + 0.1748\alpha - 0.09038\alpha^2 + 0.01652\alpha^3$$

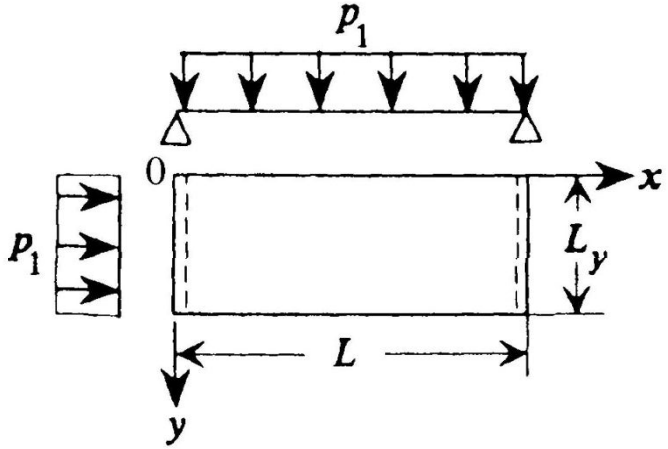
$$c_3 = 0.0009449 + 0.04083\alpha - 0.0212\alpha^2 + 0.001949\alpha^3$$

$$c_4 = -0.03425 + 0.1083\alpha - 0.02085\alpha^2 - 0.002018\alpha^3$$

$$c_5 = 0.4247 + 0.002192\alpha + 0.01065\alpha^2 - 0.003921\alpha^3$$

11.

Two edges simply supported, two edges free, uniform loading



$$(w)_{\substack{x=L/2 \\ y=L_y/2}} = c_1 p_1 L^4/D$$

$$(w)_{\substack{x=L/2 \\ y=0,L_y}} = c_2 p_1 L^4/D$$

$$(M_x)_{\substack{x=L/2 \\ y=L_y/2}} = c_3 p_1 L^2$$

$$(M_y)_{\substack{x=L/2 \\ y=L_y/2}} = c_4 p_1 L^2$$

$$(M_x)_{\substack{x=L/2 \\ y=0,L_y}} = c_5 p_1 L^2$$

$\nu = 0.3$

CONSTANTS:

$0 \le \beta \le 2.0$

$c_1 = 0.01302 - 0.0007094\beta + 0.001017\beta^2 - 0.0002372\beta^3$

$c_2 = 0.01522 + 0.0000742\beta - 0.0001711\beta^2 - 0.000033\beta^3$

$c_3 = 0.125 - 0.003249\beta + 0.0002489\beta^2 + 0.0005003\beta^3$

$c_4 = 0.0375 + 0.01038\beta - 0.02955\beta^2 + 0.008767\beta^3$

$c_5 = 0.133 + 0.0006847\beta - 0.001652\beta^2 - 0.0002326\beta^3$

TABLE 18-17 RESPONSE OF RECTANGULAR PLATES WITH FOUR SIDES SIMPLY SUPPORTED

Notation

w = deflection
θ, θ_y = slopes about lines parallel to y and x directions
M, M_y, M_{xy} = bending moments per unit length on planes normal to x and y directions and twisting moment per unit length
V, V_y = equivalent shear forces per unit length acting on planes normal to x and y axes
L, L_y = length of plate in x and y directions
h = thickness of plate
E = modulus of elasticity
ν = Poisson's ratio

$$D = \frac{Eh^3}{12(1-\nu^2)} \qquad K_{mn} = \frac{a_{mn}}{D\pi^4[(n^2/L^2) + (m^2/L_y^2)]^2} \qquad n, m = 1, 2, 3, \ldots$$

The parameters a_{mn} are given for various loadings.

General Response Expressions

1. Deflection:

$$w = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} K_{mn} \sin\frac{n\pi x}{L} \sin\frac{m\pi y}{L_y}$$

2. Slopes:

$$\theta = -\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} K_{mn} \frac{n\pi}{L} \cos\frac{n\pi x}{L} \sin\frac{m\pi y}{L_y}$$

$$\theta_y = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} K_{mn} \frac{m\pi}{L_y} \sin\frac{n\pi x}{L} \cos\frac{m\pi y}{L_y}$$

3. Bending moments:

$$M = \pi^2 D \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} K_{mn} \left[\left(\frac{n}{L}\right)^2 + \nu\left(\frac{m}{L_y}\right)^2\right] \sin\frac{n\pi x}{L} \sin\frac{m\pi y}{L_y}$$

$$M_y = \pi^2 D \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} K_{mn} \left[\left(\frac{m}{L_y}\right)^2 + \nu\left(\frac{n}{L}\right)^2\right] \sin\frac{n\pi x}{L} \sin\frac{m\pi y}{L_y}$$

4. Twisting moment:

$$M_{xy} = \pi^2 D(1-\nu) \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} K_{mn} \frac{mn}{L_y L} \cos\frac{n\pi x}{L} \cos\frac{m\pi y}{L_y}$$

5. Shear forces:

$$V = \pi^3 D \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} K_{mn} \times \left[\left(\frac{n}{L}\right)^3 + (2-\nu)\frac{n}{L}\left(\frac{m}{L_y}\right)^2\right] \cos\frac{n\pi x}{L} \sin\frac{m\pi y}{L_y}$$

$$V_y = \pi^3 D \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} K_{mn} \times \left[\left(\frac{m}{L_y}\right)^3 + (2-\nu)\frac{m}{L_y}\left(\frac{n}{L}\right)^2\right] \sin\frac{n\pi x}{L} \cos\frac{m\pi y}{L_y}$$

TABLE 18-17 (continued) RESPONSE OF RECTANGULAR PLATES WITH FOUR SIDES SIMPLY SUPPORTED

Parameters a_{mn} for Various Loadings

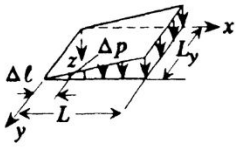
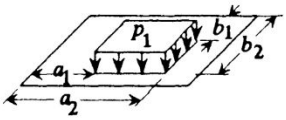
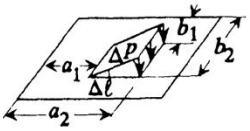
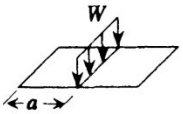
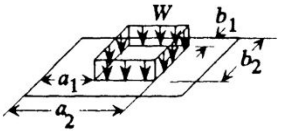
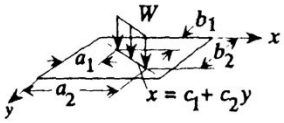
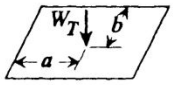
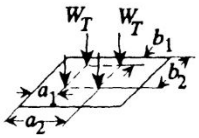
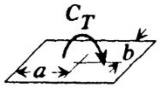
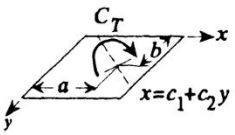
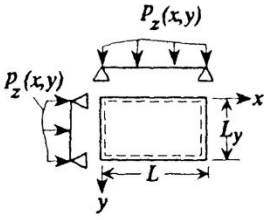
Loading	Parameter
1. Uniform load p_1 over whole plate	$a_{mn} = \dfrac{16p_1}{\pi^2 mn} \quad n, m = 1, 3, 5, \ldots$
2. Linearly varying load 	$a_{mn} = (-1)^n \dfrac{8L}{mn\pi}\dfrac{\Delta p}{\Delta \ell} \quad m = 1, 3, 5, \ldots \quad n = 1, 2, 3, 4, 5, \ldots$
3. Uniform rectangular load 	$a_{mn} = \dfrac{4p_1}{mn\pi^2}\left(\cos\dfrac{n\pi a_1}{L} - \cos\dfrac{n\pi a_2}{L}\right) \times \left(\cos\dfrac{m\pi b_1}{L_y} - \cos\dfrac{m\pi b_2}{L_y}\right)$
4. 	$a_{mn} = \dfrac{4}{nm\pi^2}\dfrac{\Delta p}{\Delta \ell}\left(\cos\dfrac{m\pi b_1}{L_y} - \cos\dfrac{m\pi b_2}{L_y}\right) \times \left[(a_1 - a_2)\cos\dfrac{n\pi a_2}{L} + \dfrac{L}{m\pi}\left(\sin\dfrac{n\pi a_2}{L} - \sin\dfrac{n\pi a_1}{L}\right)\right]$
5. Line force W (force/length) 	$a_{mn} = \dfrac{8}{\pi Lm}\sin\dfrac{n\pi a}{L}$ $m = 1, 3, 5, \ldots$ $n = 1, 2, 3, \ldots$ If this line load begins at $y = b_1$ and ends at $y = b_2$, then for a_{mn}, use $\dfrac{4W}{m\pi L}\sin\dfrac{n\pi a}{L}\left(\cos\dfrac{m\pi b_1}{L_y} - \cos\dfrac{m\pi b_2}{L_y}\right)$
6. Line force W 	$a_{mn} = \dfrac{4W}{LL_y}\left[\dfrac{L_y}{m\pi}\left(\sin\dfrac{n\pi a_1}{L} + \sin\dfrac{n\pi a_2}{L}\right) \times \left(\cos\dfrac{m\pi b_1}{L_y} - \cos\dfrac{m\pi b_2}{L_y}\right) + \dfrac{L}{n\pi}\left(\cos\dfrac{n\pi a_1}{L} - \cos\dfrac{n\pi a_2}{L}\right) \times \left(\sin\dfrac{m\pi b_1}{L_y} + \sin\dfrac{m\pi b_2}{L_y}\right)\right]$

TABLE 18-17 (continued) RESPONSE OF RECTANGULAR PLATES WITH FOUR SIDES SIMPLY SUPPORTED

Loading	Parameter
7. Line force W 	$a_{mn} = \frac{2W}{LL_y c_3}\left\{\left[\sin\left(c_3 b_2 - \frac{n\pi c_1}{L}\right) - \sin\left(c_3 b_1 - \frac{n\pi c_1}{L}\right)\right] - \left[\sin\left(c_4 b_2 + \frac{n\pi c_1}{L}\right) - \sin\left(c_4 b_1 + \frac{n\pi c_1}{L}\right)\right]\right\}$ $c_3 = \frac{m\pi L - n\pi c_2 L_y}{LL_y} \qquad c_4 = \frac{m\pi L + n\pi c_2 L_y}{LL_y}$
8. 	$a_{mn} = \frac{4W_T}{LL_y}\sin\frac{n\pi a}{L}\sin\frac{m\pi b}{L_y}$
9. 	$a_{mn} = \frac{4W_T}{LL_y}\left(\sin\frac{n\pi a_1}{L} + \sin\frac{n\pi a_2}{L}\right)\left(\sin\frac{m\pi b_1}{L_y} + \sin\frac{m\pi b_2}{L_y}\right)$
10. Line moment (force − length/length) 	$a_{mn} = -\frac{4nC}{mL^2}\cos\frac{n\pi a}{L}\left(\cos\frac{m\pi b_1}{L_y} - \cos\frac{m\pi b_2}{L_y}\right)$
11. Concentrated moment (force − length) 	$a_{mn} = -\frac{4nC_T}{L^2 L_y}\cos\frac{n\pi a}{L}\sin\frac{m\pi b}{L_y}$
12. 	$a_{mn} = \frac{4\pi C_T}{LL_y}\frac{\left(\frac{c_2 m}{L_y}\cos\frac{m\pi b}{L_y}\sin\frac{n\pi a}{L} + \frac{n}{L}\sin\frac{m\pi b}{L_y}\cos\frac{n\pi a}{L}\right)}{(1 + c_2^2)^{1/2}}$
13. 	$a_{mn} = \frac{4}{LL_y}\int_0^L \int_0^{L_y} p_z(x, y)\sin\frac{n\pi x}{L}\sin\frac{m\pi y}{L_y}dx\,dy$